Terrain: Your National Park Adventure Planner

Project Summary

Terrain is a web application designed to be a one stop shop for planning trips to US National Parks. Currently, a potential visitor needs to check the official NPS website for park details, a separate weather app for forecasts, another site for real time alerts, and yet another for air quality. This fragmented process is inefficient and can lead to visitors being unprepared for changing conditions or poor air quality.

Our project aims to solve this problem by aggregating data from multiple real time sources into a single, user friendly interface. Terrain will allow users to search for parks, view detailed park information, see live weather forecasts, check for up to the minute park alerts, and view the current Air Quality Index (AQI). Users will also be able to create accounts to save parks into personalized "trip plans," making it the perfect tool for aspiring adventurers and seasoned explorers alike.

Description of the Application

The core problem we want to solve is information fragmentation in outdoor trip planning. When planning a visit to a National Park, a user has to navigate multiple websites and apps to gather all the necessary information. For example, they might use the official NPS website for park hours, a weather service for the forecast, and other government sites like AirNow to check for recent advisories related to fire warnings, road closures, or poor air quality.

Terrain will solve this by creating a centralized hub. The application will allow a user to search for any National Park and instantly see a comprehensive dashboard for that location. This dashboard will include the park's description, available activities, real time weather data, any active alerts issued by the National Park Service, and the current air quality. This streamlines the planning process, ensuring users have the most current information to plan a safe and enjoyable trip.

Creative Component

Our project idea is to build an interactive park dashboard that brings together live data from different sources. The goal is to make it easy for users to explore a park and see important information all in one place. When a user picks a park, our app will grab data at the same time from both the National Park Service (NPS) API and a weather API.

Here’s what the dashboard will include:

- **Interactive Map:** Using a tool like Leaflet.js, we’ll show the exact location of the park on a map. It will also display visitor centers, trails, and campgrounds, using the latitude

and longitude data from the NPS API. This makes it easy for users to see what's inside the park and where everything is located.
- **Weather Widget:** We'll add a section that shows the current weather at the park, plus a 5 day forecast. It will also highlight severe weather warnings like storms, floods, or high winds, using information from the weather API. This way, people can check if it's safe or a good time to visit.
- **Live Alerts Feed:** This will display real time alerts from the NPS API, such as road closures, bear sightings, or trail hazards. It makes the dashboard more practical, since visitors can know about safety concerns before they go.
- **Air Quality Widget:** This section will show the current air quality index (AQI) for the park and explain whether the air is good, moderate, or unhealthy. A color coded Air Quality Index (AQI) badge that displays the current air quality rating (e.g., Good, Moderate, Unhealthy) and the primary pollutant, pulled from the air quality API.

This project is more challenging than just embedding a map or showing static info. We'll need to handle multiple APIs at the same time, make sure the data loads smoothly, and combine everything into one page without it feeling confusing. We'll also have to design the interface so it looks good on different devices and is easy to use. By doing this, our dashboard will create a unique, interactive experience that shows the park's location, conditions, and updates all in one place.

Usefulness

Terrain is useful because it consolidates trip planning information into a single, convenient application which doesn't exist already.

Basic Functions:

- Users can search and browse a list of all National Parks.
- Users can view a detailed page for each park, including its description, images, hours, and entry fees.
- Registered users can create an account, log in, and save parks to a personalized "Trip Plan" list.
- Registered users can create, view, update, and delete their trip plans.

Other Features:

- The interactive park dashboard with integrated live weather, air quality, and NPS alerts.
- A search function that allows users to filter parks by state or by available activities.

Comparison to Existing Apps:

- NPS Official App: This app is a great resource but lacks detailed weather forecasting.

- Weather/AQI Apps (e.g., The Weather Channel, IQAir): These provide environmental data but have no information about park activities, hours, or official alerts.
- AllTrails: This app is excellent for trail specific information but doesn't provide general park data like visitor center hours or road closure alerts.

Terrain is different because its primary value is the integration of these three distinct types of information (park info, weather, and alerts).

Realness (Data Sources)

Internal Databases

These are the relational databases we will create and maintain to handle user accounts and user generated content.

1. Data Source 1: Users Database
    - Description: This database will store all information related to user accounts. It's essential for handling authentication, authorization, and personalizing the user experience.
    - Information Captured: It will contain a users table with columns like user_id (primary key), username, email, and hashed_password.
2. Data Source 2: Trip Plans & Reviews Database
    - Description: This database will manage all user-generated content, primarily the trip plans and the personal notes or reviews users attach to parks. This database will involve creating relationships between users and parks.
    - Information Captured: It will include tables like trip_plans (to organize saved parks), a trip_plan_parks join table (to link users' plans to specific parks from the NPS dataset), and a reviews table where users can store notes or ratings for parks they've visited. This table will have foreign keys linking back to both the users and parks tables.

External Databases

We will use at least two real time, high quality data sources for this project.

1. Data Source 1: The National Park Service
    - Where it's from: https://www.kaggle.com/datasets/thedevastator/the-united-states-national-parks
    - Format: JSON.
    - Data Size: The API provides access to data on over 400 national parks. The park's endpoint alone contains thousands of data points (descriptions, locations, activities, images, etc.), easily satisfying the greater than 1K data requirement.

    - Information Captured: Park names, descriptions, locations (latitude/longitude), images, operating hours, entrance fees, contact info, available activities, visitor center locations, campground information, and real time alerts.
2. Data Source 2: U.S. Travel Destinations Dataset
    - Where it's from: https://www.kaggle.com/datasets/mosahle7/us-travel-destinations/data
    - Format: We will load the data from the provided CSV file into a new destinations table in our database.
    - Information Captured: This dataset includes a wide range of U.S. destinations with information on top attractions, recommended travel times, average daily cost, and user ratings.
    - How it will be used: This data will enrich the trip planning experience by showing users other points of interest and attractions near the National Parks they are viewing. It will also allow us to provide helpful budget estimates for their trip.
3. Data Source 3: The OpenWeatherMap One Call API
    - Where it's from: OpenWeatherMap, a popular weather data provider. They offer a generous free tier for developers.
    - Format: JSON.
    - Data Size: This is a real time service. We will make requests for specific locations as needed.
    - Information Captured: Current weather conditions (temperature, wind, humidity), hourly forecasts for 48 hours, and daily forecasts for 8 days. This API is called using the latitude and longitude we get from the NPS API for each park.
4. Data Source 4: The AirNow API
    - Where it's from: The U.S. Environmental Protection Agency (EPA). It provides real time air quality data for locations across the United States.
    - Format: JSON.
    - Information Captured: The current Air Quality Index (AQI), the primary pollutant (e.g., Ozone, PM2.5), and the associated health level category. This API will also be called using the park's coordinates.

Functionality (User Interaction)

The application will support full CRUD (Create, Read, Update, Delete) operations for registered users.

- Who: An authenticated user.
- What/When (Create):
    - A user will submit a registration form with a username, email, and password to create a new account.
    - After logging in, a user can click an "Add to Trip" button on a park's page. This will create a new entry in their "Trip Plan," linking their user ID to the park ID.

    - A user can add personal notes to a park within their trip plan.
- Who: Any visitor (logged in or not).
- What/When (Read):
    - A visitor can read (view) a list of all parks on the homepage.
    - A visitor can use a search bar or filters to search for specific parks and read the results.
    - A visitor can click on a park to read its detailed information page, including the integrated weather and alerts.
- Who: An authenticated user.
- What/When (Update):
    - On their profile page, a user can view a trip plan and click an "edit" button next to their personal notes. This will allow them to update the text of their notes for that park.
- Who: An authenticated user.
- What/When (Delete):
    - On their profile page, a user can click a "Remove" button next to a park in their trip plan. This will delete that park from their plan.
    - A user could also have an option to delete their entire trip plan.

In addition to the CRUD operations, our application will feature a search and filtering system designed to help users discover new parks that perfectly match their interests and travel plans.

The search functionality will be implemented for the following:

- Search by Park Name: The search will be dynamic, suggesting parks that match the user's input as they type, making it fast and intuitive.
- Dynamic Results Display: The search results will be displayed as a grid of visually appealing park cards. Each card will feature a primary image of the park, its name, and its state. As the user types in the search bar or applies filters, the grid of results will update in real time without requiring a page reload, creating a smooth and responsive experience. Clicking on any park card will take the user directly to that park's detailed information page.

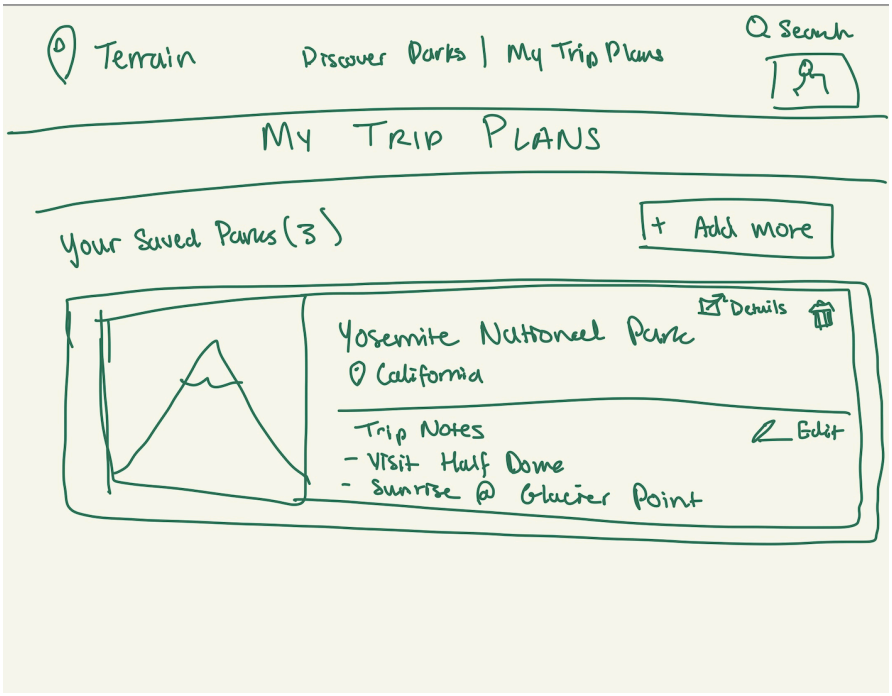
Low Fidelity UI Mockup

1. Homepage: Features a prominent search bar and a grid of beautiful photos representing featured parks.
2. Park Details Page: A large banner image of the park, below that the park's name and location. Underneath that there will be three clear tabs:
    - About: Contains the park description, hours, fees, and activities.

- Weather: Shows the dynamic weather widget with the 5 day forecast.
- Live Alerts: Displays a feed of the latest official NPS alerts.

3. User Profile Page: A simple dashboard titled "My Trip Plans" that shows a list of saved parks.





Project Work Distribution

Our team will distribute the backend and frontend tasks as follows:

- Devanshi Sawhney - ds63 (Frontend):
    - Responsible for the overall UI/UX design and implementation.
    - Will build the React/Vue components for the Homepage, Park Details Page, and User Profile.
    - Will handle CSS and ensure the application is responsive.
- Aditi Shukla - shukla8 (Backend - NPS API & Database):
    - Responsible for setting up the main Node.js/Express server.
    - Will write the server side logic to fetch data from the NPS API.
    - Will design the PostgreSQL database schema for parks and activities.
    - Will create the backend routes (/api/parks, /api/parks/:id).
- Hetvi Patel - hetvip2 (Backend - User Authentication & Database):
    - Responsible for implementing user registration and login functionality (e.g., using Passport.js).
    - Will design the database schema for users and their saved "Trip Plans" (the join table between users and parks).
    - Will create the secure backend routes for all user specific actions (CRUD for trip plans).
- Manan Vora - mananv2 (Creative Components & Integration):
    - Responsible for integrating the OpenWeatherMap API.
    - Will work with Devanshi to build the interactive map and weather widget.
    - Will focus on the backend logic for the dashboard page that combines data from the NPS and weather APIs in a single response to the front end.